Glucose Levels Across Maternity

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LIST OF ABBREVIATIONS

ABBREVIATION	DEFINITION
MD	Maternal dysglycemia
APO	Adverse pregnancy outcomes
GDM	Gestational diabetes mellitus
OGTT	Oral glucose tolerance test
BMI	Body mass index
AUC	Area Under the Curve
CRF	Case Report Form
CGM	Continuous Glucose Monitoring
GCP	Good Clinical Practice
HbA1c	Hemoglobin A1c
IQR	Interquartile Range
TIR	Time in range
QA	Quality Assurance
QC	Quality Control
SMBG	Self-Monitoring of Blood Glucose
TSD	Total standard deviation
CONGA	Continuous overlapping net glycemic action

PROTOCOL SUMMARY

PARTICIPANT AREA	DESCRIPTION
Title	Glucose Levels Across Maternity (GLAM)
Précis	Expectant mothers will wear a blinded CGM periodically or continuously during their pregnancy while receiving routine prenatal care to assess the frequency of CGM-measured dysglycemia, the association of CGM-measured dysglycemia with oral glucose tolerance test results, and the association of CGM-measured dysglycemia with adverse pregnancy outcomes.
Device	Commercially- available Dexcom G6 Pro or ProQ
Objectives	1) Characterize glycemic patterns in healthy pregnant women using CGM data; 2) Characterize glycemic patterns to detect or identify if certain glycemic outcomes, overall, or at different time points, are more indicative of an adverse pregnancy outcome (APO). 3) Explore the extent to which CGM-derived glycemic patterns throughout pregnancy and GDM diagnosis per standard of care correlate with adverse pregnancy outcomes (APOs).
Study Design	Observational, nonintervention study
Number of Sites	2
Study Data Collection/Endpoints	• Electronic medical record (EMR) data, including patient health history, demographics, diagnoses, co-morbid conditions, medications, procedures, etc. • CGM glucose data • Meal time marker recording completed by participants who agree to collect data via a phone app • Results of Oral Glucose Tolerance Test (OGTT) • APOs of interest in mother and child • Infant characteristics at delivery, including weight and length
Population and Eligibility Criteria	Women with singleton pregnancies seeking routine prenatal care at or before gestational time of 16 weeks, 6 days. Inclusion Criteria: • Singleton pregnancy with viability verified by 12 weeks gestation on ultrasound* • Maternal age of 18 years and older • Gestational age up to 16 weeks, 6 day determined on ultrasound* • English speaking • HbA1c <6.5%** Exclusion Criteria: • Signs of abnormalities in fetal or placental development at first routine prenatal visit/ultrasound* • Planned termination of pregnancy or any indications of miscarriage • Prior gastric bypass surgery

PARTICIPANT AREA	DESCRIPTION
	• Pregravid diabetes (type 1 or type 2) • Unwillingness/inability to wear Dexcom CGM device • Unwillingness to attend routine antenatal obstetric appointments • Use of oral systemic steroids • Use of any medication intended to lower blood glucose, including insulin and non-insulin therapies, at any point during the pregnancy, up to and including at the time of enrollment. • Deemed unable to participate for medical reasons identified by study staff **although it is preferable that ultrasound results be available prior to enrollment, if ultrasound results are not available at the time of enrollment, participant can be enrolled and have CGM initiated but will be dropped if not eligible after results are available* ***if HbA1c result not available at time of enrollment, participant can be enrolled and have CGM initiated but will be dropped if not eligible after results are available*
Sample Size	~800
Participant Duration	Variable: from the time of enrollment until the end of pregnancy
Risks	As an observational, nonintervention study following standard care, risks are minimal. There is the possibility of discomfort and skin reaction from the CGM sensor. There are no known risks to the fetus. (see section 1.2.1)
Benefits	There may be no direct benefit to the participant from participating in this study. Participants will be provided with a report of their CGM data after the conclusion of the study, which may be useful for their glycemia management in future pregnancies. By helping to establish a normal range of sensor glucose values in pregnant women, participation in this study is an important contribution to the development of a better understanding of glucose metabolism during pregnancy.
Protocol Overview/Synopsis	Expectant mothers will be enrolled by 16 weeks, 6 days of the gestational period. Informed consent will be obtained, and eligibility assessed. A blinded Dexcom G6 Pro or ProQ sensor will be placed, and the participant will be instructed on its care and mailing of the transmitter to the coordinating center after 10 days. The participants may also return transmitters to the study site. A new sensor may be placed at each standard care office visit and/or at home. Training of sensor insertion may be performed through virtual methods (i.e. video conferencing) remotely or in-person. Participants will be encouraged to insert new sensors at home between visits, but this will be optional. As per usual care, an OGTT will be performed between 24-28 weeks. At conclusion of pregnancy, data will be recorded for APOs.

1 Background Information

1.1 Introduction

Maternal dysglycemia (MD) is a broad term used to refer to abnormal patterns in plasma glucose concentrations, indicative of impaired glycemic homeostasis, occurring during the gestational period. MD is often a result of pregestational diabetes mellitus (Type 1 and Type 2 diabetes present before pregnancy) or gestational diabetes mellitus (GDM), and it is well documented that dysglycemia in this population is associated with increased risk of adverse pregnancy outcomes (APOs).[1–3] However, a growing body of evidence suggests that MD may be present in women without a diabetes diagnosis during pregnancy and that this MD is similarly associated with increased APO risk in this population.[4]

Glycemic patterns naturally change through pregnancy as insulin sensitivity progressively reduces across gestation.[3,5] However, a clear definition of healthy glucose patterns during pregnancy is currently lacking, with the most uncertainty around glycemic profiles in nondiabetic pregnant women. As such, many questions remain regarding how patterns in glucose levels during gestation impact pregnancy outcomes.

Currently, MD is generally discovered only through standard-of-care screening and diagnostic tests for GDM. Diagnostic tests include an oral glucose tolerance test (OGTT) administered between the 24th and 28th week of gestation.[5] Risk assessment and diagnostic criteria for GDM with the OGTT differ across organizations and there is no universal standard.[6] Researchers have found significant variability in the results of OGTT tests, highlighting an additional weakness in the current standard of care. This variability is especially pronounced when plasma glucose concentrations are near the bound between normoglycemia and hyperglycemia. Retesting is thus recommended for borderline OGTT results, especially for those considered to be at risk for GDM due to high BMI, previous GDM, or family history of diabetes.

Recent findings suggest that MD and its correlated increased risk with many APOs is best mitigated with early detection and intervention prior to the 24th week of gestation, particularly in the first trimester or the beginning of the second trimester.[6] This underscores another potential weakness in the current standard of care in which the OGTT is administered towards the end of the second trimester when- as an increasing body of clinical research suggests- it may be too late to counteract the adverse effects of MD. Also, growing evidence supports that maternal hyperglycemia below the OGTT-thresholds used to diagnose GDM is associated with increased risk of maternal and neonatal adverse outcomes.[4] As a result of inconsistent diagnostic criteria and issues surrounding the timing and reproducibility of the results, some pregnant women may not receive a GDM diagnosis while still being at heightened risk of MD and APOs.[4]

Several APOs have been shown in the literature to be associated with MD, even below the diagnostic threshold for GDM. These include birth weight above the 90th percentile for gestational age (macrosomia), cesarean delivery, clinically diagnosed neonatal hypoglycemia,

cord-blood serum C-peptide level above the 90th percentile, preterm birth (before 37 weeks gestation), shoulder dystocia or birth injury, need for intensive neonatal care, and preeclampsia.[4] Infants are also at higher risk for obesity and diabetes later in life.[7] These APOs appear to be caused by the excess glucose passing through the placenta, which triggers "fetal pancreatic beta-cell production of insulin to control fetal blood glucose levels."[8] Insulin in the fetus acts as a growth hormone which leads to the increase in fetal size.

In the current study, we propose that use of continuous glucose monitoring (CGM) device during pregnancy may be an improved tool for timely detection of MD. Specifically, we hypothesize that we can identify glycemic irregularities using CGM data collected from the first routine prenatal visit to delivery and that CGM-derived metrics will be correlated with adverse outcomes.

Towards the goal of assessing these hypotheses, we aim to:

1. Characterize glycemic patterns in healthy pregnant women using CGM data;
2. Characterize glycemic patterns to detect or identify if certain glycemic outcomes, overall, or at different time points, are more indicative of an adverse pregnancy outcome (APO).
3. Explore the extent to which CGM-derived glycemic patterns throughout pregnancy and GDM diagnosis per standard of care correlate with adverse pregnancy outcomes (APOs).

These objectives are not fully addressed and not well understood in the existing literature. The majority of research regarding CGM use during pregnancy has been centered around women with pregestational diabetes and women diagnosed with GDM by an OGTT in the second half of pregnancy. There is also limited information about maternal glucose levels throughout the entire course of gestation, with some studies using 72-hour CGM in the third trimester.[9] This research has found that CGM use is safe and accurate in pregnant populations;[9–11] it was also found that CGM use increases detection of dysglycemic events and their clinical treatment compared with self-monitoring blood glucose (SMBG) measurements.[9] These previous studies used older CGM models, which were less accurate than the current CGMs on the market. Thus, we intend to address a long-standing research need and pursue the goal of transforming the most state of the art CGM data into actionable information.

This study is expected to be the first step in a multi-phase process. The findings from this study will provide critical information for the design of subsequent studies, where we intend to investigate the use of unblinded CGM devices as an intervention and engagement tool during pregnancy and the impact that awareness of gestational glucose levels has on risk of APOs and later childhood development and health outcomes.

1.2 Potential Risks and Benefits

1.2.1 Known Potential Risks

The CGM sensor may briefly produce pain when it is inserted just underneath the skin. There is a low risk for developing a local skin infection at the site of the sensor placement. Itchiness, redness, bleeding, and bruising at the insertion site may occur as well as local tape reactions. On

rare occasions, the sensor may break and leave a small portion of the sensor under the skin that may cause redness, swelling or pain at the insertion site.

In any study with transmission of data, there is always a risk of breach of confidentiality. However, all electronic data transfers will be encrypted, and data will be stored securely to prevent such as breach.

1.2.2 Known Potential Benefits

There will be no direct benefit to the participant from participating in this study. By helping to establish a normal range of sensor glucose values in pregnant women, participation in this study is an important contribution to the development of a better understanding of glucose metabolism during pregnancy. Participation in this study could improve prenatal care of women in the future.

1.2.3 Risk Assessment

The study is considered *minimal risk* as defined in 45CFR46.102 since the probability and magnitude of harm or discomfort anticipated in the research are not greater, in and of themselves, than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests.

1.3 General Considerations

The study is being conducted in compliance with ethical principles that have their origin in the Declaration of Helsinki, with the protocol described herein, and with the standards of Good Clinical Practice (GCP).

When feasible, data will be directly collected in electronic case report forms, which will be considered the source data.

2 Study Protocol

2.1 Participant Recruitment and Enrollment

Enrollment will proceed with the goal of 800 participants commencing the study (i.e., determined to be eligible and initiate CGM wear). A maximum of 1,000 individuals may be enrolled for eligibility screening (i.e., electronically obtain informed consent) in order to achieve this goal.

Study participants will be recruited/enrolled at two clinical centers in the U.S. at a routine prenatal visit or via a remote (i.e. phone or video conferencing) process. All eligible participants will be included without regard to gender, race, or ethnicity. There is no absolute restriction on the number of participants to be enrolled by each site toward the overall recruitment goal, but it is expected that each clinical center will have approximately 400 participants.

2.1.1 Informed Consent and Authorization Procedures

Potential eligibility will be assessed as part of a routine-care examination. Before completing any procedures or collecting any data that are not part of usual care, the informed consent process approved by the JCHR IRB will be followed.

The study protocol will be discussed with the potential study participant by study staff either on site or remotely. The potential study participant will be given the Informed Consent Form to read (either in person, by mail or by email). Potential study participants will be encouraged to discuss the study with family members and their personal physicians(s) before deciding whether to participate in the study. If the person decides to participate in the study, the site staff member authorized to obtain verbal consent will use the Obtain an ID link on the Study Website to electronically document the participant's verbal consent. The Obtain an ID form will document the following consent steps have been completed:

1. The participant has read and understands the current informed consent form.
2. The study staff member conducting consent has explained how the participant's personal information will be collected and used for this study, and that this is necessary to be in the study.
3. The participant's questions and concerns have been addressed.
4. The participant has been reminded that the study is voluntary.
5. A copy of the consent has been made available to the participant for her records.
6. The participant has confirmed that she would like to be in the study at this time.
7. The participant wants to be in the optional portion of the study and receive text messages (yes / no).
8. The participant wants to be in the optional portion of the study and log meals (yes / no).

Next, the study staff member will submit the electronic form and the study website will generate a four-digit verification code, which will be given to the participant verbally. Once the participant has the verification code, the staff member will email them the consent confirmation link to the website where the verification code must be entered. The staff member will ask the

participant to check their email for the link and enter the code accordingly to verify that they have verbally consented and are agreeing to participant in the study. If the participant cannot access their email in-clinic, they can allow the site to pull up the link on a site tablet/computer and enter the verification code there. The participant will still have record of the link in their email, but will not have to re-enter the verification code again.

As part of the informed consent process and per HIPAA (Health Insurance Portability and Accountability Act) standards, each participant will be asked to verbally provide HIPAA authorization for release of personal information. The investigator, or his or her designee, will review the study-specific information that will be collected and to whom that information will be disclosed as part of the consent process. After speaking with the participant, questions and concerns will be addressed regarding authorization.

A participant is considered enrolled when the informed consent process is completed, verbal consent has been attained from the participant, the process has been recorded electronically, and the participant has entered their verification code on the website. For more information, please see the Informed Consent Process Guide.

2.2 Eligibility Criteria

2.2.1 Participant Inclusion Criteria

Individuals must meet all of the following inclusion criteria in order to be eligible to participate in the study.

1) Singleton pregnancy with viability verified by 12 weeks gestation on ultrasound*
2) Maternal age of 18 years and older
3) Gestational age up to 16 weeks, 6 days determined on ultrasound*
4) English speaking
5) HbA1c <6.5%**

2.2.2 Participant Exclusion Criteria

Individuals meeting any of the following exclusion criteria at baseline will be excluded from study participation.

1) Signs of abnormalities in fetal or placental development at first routine prenatal visit/ultrasound*
2) Planned termination of pregnancy or any indications of miscarriage
3) Prior gastric bypass surgery
4) Pregravid diabetes (type 1 or type 2)
5) Unwillingness/inability to wear Dexcom CGM device
6) Unwillingness to attend routine antenatal obstetric appointments
7) Use of oral systemic steroids

8) Use of any medication intended to lower blood glucose, including insulin and non-insulin therapies, at any point during the pregnancy, up to and including at the time of enrollment.

9) Deemed unable to participate for medical reasons identified by their physician

**although it is preferable that ultrasound results be available prior to enrollment, if ultrasound results are not available at the time of enrollment, participant can be enrolled and have CGM initiated but will be dropped if not eligible after results are available*

***if HbA1c result not available at time of enrollment, participant can be enrolled and have CGM initiated but will be dropped if not eligible after results are available*

2.3 Data Collection and Testing

Except for the blinded CGM data (see section 2.4) and meal marker data (see section 2.5), no data will be collected, or procedures performed that are not part of usual care and not being recorded in the electronic medical record.

These data will include patient health history, prior pregnancies, demographics, medical conditions, medications, medical procedures, height/weight, blood pressure, relevant physical examination abnormalities, HbA1c, relevant laboratory abnormalities, and medical conditions developing during the pregnancy.

2.4 CGM

The CGM used in the study will be a commercially-available, unmodified Dexcom G6 Pro and Dexcom ProQ. Each CGM sensor will be worn for up to 10 days on the skin. The location of sensor insertion can be abdominal, upper buttocks, or extremity.

At the enrollment visit, a CGM sensor will be inserted and the participant will be instructed on its use and care. If the participant does not come into the clinic for the enrollment visit, they may be trained on CGM insertion via a video conference. The device will be blinded so that the participant is not able to see the sensor readings. The sensor will be worn for up to 10 days. A new CGM sensor may be inserted at clinic visits if the participant is not inserting sensors at home and the participant will be reminded of the use instructions.

Participants will be encouraged to place additional sensors at home between visits, but this will be optional if they are coming into clinic for regular prenatal visits.

CGM will continue through labor and delivery. The sensor and transmitter will be collected from the participants 1 or 2 days postpartum prior to discharge to home.

Participants will be provided with postage-paid envelopes to send the CGM transmitters to the Coordinating Center at the end of the sensor wear (after 10 days or sooner if the sensor becomes dislodged).

A report summarizing the CGM data will be provided to the participant to share with her health care provider after the study is completed.

2.5 Recording Time of Meals

Participants who consent to recording of timing of meals, will record the timing of meals using a phone app for 1 week during the 18-22 week and 32-34 week time periods of CGM wear.

2.6 Office Visits

All visits will follow a standard care schedule. Some of these visits may be done via telemedicine. There will be no study-specific data collection that is not recorded in the EMR. Selected data will be entered into the study database on the study website.

2.7 Oral Glucose Tolerance Test

An OGTT, which may involve more than one test, will be performed as part of usual care, according to the sites' usual routine, between 24 and 28 weeks. Results will be recorded in the study database.

CGM will be worn during the OGTT unless this is not feasible (e.g., timing related to new sensor placement).

2.8 Reportable Adverse Events

For this protocol, a reportable adverse event includes an adverse device effect (ADE) that requires medical treatment, such as a skin infection at a sensor location; or a sensor needle fracture with a portion of the sensor needle retained beneath the skin.

2.9 Data Collected at End of Pregnancy

After completion of the pregnancy, data to be entered into the study database will include gestation week at time of delivery/end of pregnancy, maternal complications, fetal complications, infant birth weight and height.

2.10 Safety Oversight

As this is a minimal risk nonintervention, observational study, there will not be a Medical Monitor or Data and Safety Monitoring Board. All reported adverse events will be reviewed by the Coordinating Center staff and when applicable discussed with the Study Chair.

3 Miscellaneous Considerations

3.1 Unanticipated Problems

Site investigators will promptly report to the Coordinating Center on an eCRF all unanticipated problems meeting the criteria below. For this protocol, an unanticipated problem is an incident, experience, or outcome that meets all of the following criteria:

- Unexpected (in terms of nature, severity, or frequency) given (a) the research procedures that are described in the protocol related documents, such as the IRB-approved research protocol and informed consent document; and (b) the characteristics of the subject population being studied
- Related or possibly related to participation in the research (possibly related means there is a reasonable possibility that the incident, experience, or outcome may have been caused by the procedures involved in the research)
- Suggests that the research places participants or others at a greater risk of harm than was previously known or recognized (including physical, psychological, economic, or social harm)

3.2 Participant Compensation

Participant compensation will be specified in the informed consent form.

3.3 Participant Withdrawal

Participation in the study is voluntary, and a participant may withdraw at any time. For participants who withdraw, their data will be used up until the time of withdrawal.

The site will withdraw the participant if (1) the HbA1c result, which was not available at the time of enrollment, is $\geq$6.5%, (2) ultrasound results, which were not available at the time of enrollment, does not indicate a viable fetus, (3) pregnancy terminates before the planned end of study participation.

3.4 Confidentiality

For security and confidentiality purposes, participants will be assigned an identifier that will be used instead of their name. Protected health information gathered for this study will be shared with the coordinating center, the Jaeb Center for Health Research in Tampa, FL. De-identified participant information may also be provided to research sites involved in the study.

4 Statistical Considerations

4.1 Statistical and Analytical Plans

The approach to sample size and statistical analyses are summarized below and will detailed further in a Statistical Analysis Plan.

4.1.1 Overview

We established power around hypothesized improvements in sensitivity of a CGM-based risk assessor at determining risk of APOs relative to those of the OGTT.

We used data collected from previous internal research on a cohort of over 300 pregnant women, as well as historic data from the University of Pennsylvania, to inform rates of GDM (i.e. positive OGTT) and APOs that we expect in phase 1 of this study. Our current list of APOs with corresponding prevalence listed in parenthesis includes:

- C-section (28%)
- Large for gestational age [LGA] (12%)
- Preterm birth (12%)
- Preeclampsia (12%)

GDM was diagnosed in 8.4% of the mothers in this study with the OGTT and has varying association with the APOs of interest. Power calculations were conducted for these APOs individually.

4.1.2 Assumptions

Two diagnostic tests, CGM-risk assessor and OGTT assessor, will be administered to each of the N pregnant women with each test predicting whether the APO of interest will occur. Of those N pregnant women, N_{APO} will have the APO of interest and the following 2-by-2 table will be constructed to compare the two diagnostic sensitivities:

		CGM		
		Positive	Negative	
OGTT	Positive	a	b	a+b
	Negative	c	d	c+d
		a+c	b+d	N_{APO}

The McNemar test will be used to test whether the two sensitivities are different, which only requires knowledge of the number of discordant pairs: b and c. Therefore, the sample size is based on the P(b), P(c), and P(N_{APO}) for each APO.

The following accuracies of OGTT and different APOs were taken from the historical data:

Table 1: Breakdown of mothers by APO and OGTT from historical data

APO	Maternal Breakdown by Subgroup*	
	APO and Positive OGTT	**APO and Negative OGTT**
C-Section	4%	24%
LGA	2%	10%
Preeclampsia	1%	11%
Preterm birth	1%	11%

* APO prevalence rate is the sum of the two columns.

We assume the OGTT and CGM tests are dependent. Specifically, for pregnant women with an APO and a positive OGTT (i.e., OGTT predicting APO to occur), we assume the CGM will be able to positively predict APO for 90-99% of these women. For pregnant women with an APO and a negative OGTT, we assume the CGM will be able to positively predict APO for 15-20% of these women. The APO prevalence rate and sensitivity of OGTT in Table 1 combined with the conditional probabilities assumed are used to calculate the discordant pair probabilities.

4.1.3 Sample Size Determination

The required sample size to detect the difference between the OGTT and CGM-risk assessor sensitivity from a two-sided McNemar test with an α-level of 0.05 is calculated using the following formula[12] :

$$N = \frac{\left(z_{1-\frac{\alpha}{2}} \left(\frac{P(b)}{P(c)} + 1 \right) + z_{1-\beta} \sqrt{\left(\frac{P(b)}{P(c)} + 1 \right)^2 - \left(\frac{P(b)}{P(c)} - 1 \right)^2 (P(b) + P(c))} \right)^2}{P(N_{APO}) \left(\frac{P(b)}{P(c)} + 1 \right)^2 (P(b) + P(c))}$$

where N is the required sample size to achieve 1-β power with a type 1 error rate of α.

The following table summarizes the assumed conditional probabilities between OGTT and CGM risk assessors, their sensitivities, and sample sizes needed to achieve 80% and 90% power to detect improvement in sensitivity.

Table 2. Required Sample Sizes for Different Assumptions

APO	Prevalence Rate	Conditional Probability of Positive CGM-Risk Assessor		OGTT Sensitivity	CGM Risk Assessor Sensitivity	Sample Size	
		Given APO and Positive OGTT	Given APO and Negative OGTT			80% Power	90% Power
C-Section	28%	99%	20%	14%	31%	159	210

	28%	95%	18%	14%	29%	201	265
	28%	90%	15%	14%	26%	299	396
LGA	12%	99%	20%	17%	33%	385	506
	12%	95%	18%	17%	31%	496	656
	12%	90%	15%	17%	28%	770	1022
Preeclampsia	12%	99%	20%	8%	27%	342	449
	12%	95%	18%	8%	24%	408	538
	12%	90%	15%	8%	21%	552	731
Preterm Birth	12%	99%	20%	8%	27%	342	449
	12%	95%	18%	8%	24%	408	538
	12%	90%	15%	8%	21%	552	731

Using a 2-sided McNemar test for comparing two sensitivities with an α of 0.05, and assuming a conditional probability of 90% and 15% for a positive CGM risk assessor given a positive and negative OGTT, respectively, for women with an APO, the study will require a sample size of **770** to achieve at least 80% power in detecting a difference in the sensitivities of the OGTT and CGM risk assessor. This will be increased to **856** to account for a 10% dropout rate.

4.1.4 Additional Considerations

- We also considered shoulder dystocia in this power analysis but did not reach at least 80% power for this outcome with our analysis approaches due to low prevalence.
- We will require a larger sample size to correlate CGM metrics to less common individual APOs. Individual APOs whose rates do not vary greatly by GDM status will also require larger sample sizes to detect significance.

4.2 Analysis Cohort

All participants enrolled with ≥72 hours of CGM data, a definitive GDM status based on OGT testing, and known APO status will be included.

4.3 Statistical Analysis for Primary Objectives

The statistical analysis for each primary objective is detailed below.

4.3.1 Characterization of Glycemic Trends in Healthy Pregnant Women Using CGM Data

We define healthy pregnancies as those without a diagnosis of GDM per the oral glucose tolerance test and without one of our identified adverse outcomes during the course of pregnancy and through delivery.

This objective is exploratory. The following glycemic outcomes will be tabulated by participant:

- Mean glucose
- Glucose SD

- Glucose coefficient of variation
- % time in range 63-120 mg/dL
- % time in range 63 – 140 mg/dL
- % time <63 mg/dL
- Area over the curve 63 mg/dL
- % time >120 mg/dL
- % time >140 mg/dL
- Area under the curve 120 mg/dL
- Area under the curve 140 mg/dL

Mean and SD or summary statistics appropriate to the distribution will be reported. These will be calculated at the participant level as well as the trimester and 4-week levels where the distinction is available to assess the population overall and to explore changes in these metrics over the course of pregnancy. These metrics will also be summarized by daytime and nighttime.

Metrics for standardized meals include but are not limited to:

- Glucose level at 30 minutes, 1 hour, and 2 hours
- Post-prandial peak
- Time to post-prandial peak
- Glucose meal excursion (post-prandial peak glucose minus glucose prior to meal)

Other CGM metrics will be explored as detailed in the statistical analysis plan.

4.3.2 CGM Metrics vs. Unhealthy Pregnancies

An exploratory analysis will utilize the characterization of glycemic profiles of healthy pregnancies (Section 4.3.1) and unhealthy pregnancies (defined as GDM diagnosis by OGTT or having any APOs). The same glycemic characteristics from the first primary objective (Section 4.3.1) will be tabulated for mothers with and without a diagnosis of GDM and with and without APOs. Mean and SD or summary statistics appropriate to the distribution will be reported.

Supervised and unsupervised learning methods will use a subset of these CGM metrics to identify glycemic profiles that are indicative of unhealthy pregnancy outcomes, i.e. GDM or APOs.

4.3.3 Exploration of the Extent to Which CGM-Derived Glycemic Patterns Throughout Pregnancy and GDM Diagnosis per Standard of Care Correlate with APOs

Primary focus of this objective will be on increases in sensitivity with a CGM-based risk assessor over the OGTT on mothers with the APOs of interest. These will both indicate improvement in identifying APO risk. This study is not powered to measure differences in specificity, positive predictive value (PPV), and negative predictive value (NPV), but these metrics will be captured for observational comparisons between the OGTT and the CGM risk assessment method.

Supervised learning methods will use CGM-derived metrics computed throughout gestation to classify risk of individual APOs. Cross-validation will be used to evaluate the highest performing CGM-based risk assessor and model performance will be assessed via methods including the construction of the receiver operating characteristic (ROC) curves for each APO. We expect improvement over the OGTT (e.g. a diagnosis of GDM via the OGTT indicates high risk of APOs) with this CGM-based risk assessor at identifying risk of APOs. Metrics including sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) will be calculated for both the CGM risk assessor and the OGTT for each APO. For participants with the APO, McNemar's test with $\alpha=0.05$ will be run on a 2x2 contingency table of this data to determine if there are significant differences in sensitivity between the two methods in identifying APO risk.

4.4 Adherence and Retention Analysis

The following will be tabulated by participant, overall and by trimester:

- Hours or days of CGM readings
- Number of CGM sensors worn

Mean and SD or summary statistics appropriate to the distribution will be reported.

4.5 Baseline Descriptive Statistics

Baseline demographic and clinical characteristics of the study cohort will be tabulated. Mean and SD or summary statistics appropriate to the distribution will be reported.

4.6 Exploratory Analyses

Additional exploratory analyses will be described in the SAP.

5 Data Collection and Monitoring

5.1 Case Report Forms and Other Data Collection

The main study data are collected on electronic case report forms (eCRFs) and from uploads of data stored on a CGM sensor transmitter.

5.2 Study Records Retention

Each participating site will maintain appropriate medical and research records for this trial, in compliance with ICH E6 and regulatory and institutional requirements for the protection of confidentiality of participants.

Study documents should be retained for a minimum of 3 years after the completion of the study. These documents should be retained for a longer period, however, if required by local regulations.

5.3 Quality Assurance and Monitoring

Designated personnel from the Coordinating Center will be responsible for maintaining quality assurance (QA) and quality control (QC) systems to ensure that the clinical portion of the trial is conducted and data are generated, documented and reported in compliance with the protocol, Good Clinical Practice (GCP) and the applicable regulatory requirements, as well as to ensure that the rights and wellbeing of trial participants are protected and that the reported trial data are accurate, complete, and verifiable. Adverse events will be prioritized for monitoring.

Coordinating Center representatives or their designees may visit the study facilities at any time in order to maintain current and personal knowledge of the study through review of the records, comparison with source documents, observation and discussion of the conduct and progress of the study. The investigational site will provide direct access to all trial related sites, source data/documents, and reports for the purpose of monitoring and auditing by the sponsor, and inspection by local and regulatory authorities.

5.4 Protocol Deviations

A protocol deviation is any noncompliance with the clinical trial protocol, GCP, or procedure requirements. The noncompliance may be either on the part of the participant, the investigator, or the study site staff. As a result of deviations, corrective actions are to be developed by the site and implemented promptly.

The site PI/study staff is responsible for knowing and adhering to their IRB requirements. Further details about the handling of protocol deviations will be included in the monitoring plan.

6 Ethics/Protection of Human Participants

6.1 Ethical Standard

The investigator will ensure that this study is conducted in full conformity with Regulations for the Protection of Human Participants of Research codified in 45 CFR Part 46, 21 CFR Part 50, 21 CFR Part 56, and/or the ICH E6.

6.2 Institutional Review Boards

The protocol, informed consent form(s), recruitment materials, and all participant materials will be submitted to the IRB for review and approval. Approval of both the protocol and the consent form must be obtained before any participant is enrolled. Any amendment to the protocol will require review and approval by the IRB before the changes are implemented to the study. All changes to the consent form will be IRB approved; a determination will be made regarding whether previously consented participants need to be re-consented.

6.3 Informed Consent Process

6.3.1 Consent Procedures and Documentation

Informed consent is a process that is initiated prior to the individual's agreeing to participate in the study and continues throughout the individual's study participation. Extensive discussion of risks and possible benefits of participation will be provided to the participants and their families. Consent forms (written or electronic) will be IRB-approved and the participant will be asked to read and review the document. The investigator will explain the research study to the participant and answer any questions that may arise. All participants will receive a verbal explanation in terms suited to their comprehension of the purposes, procedures, and potential risks of the study and of their rights as research participants. Participants will have the opportunity to carefully review the consent form and ask questions prior to signing.

The participants should have the opportunity to discuss the study with their surrogates or think about it prior to agreeing to participate. The participant will complete the electronic informed consent page prior to any procedures being done specifically for the study. The participants may withdraw consent at any time throughout the course of the trial. A copy of the informed consent document will be given to the participants for their records. The rights and welfare of the participants will be protected by emphasizing to them that the quality of their medical care will not be adversely affected if they decline to participate in this study.

6.3.2 Participant and Data Confidentiality

Participant confidentiality is strictly held in trust by the participating investigators, their staff, and the sponsor(s) and their agents. Therefore, the study data, and all other information generated will be held in strict confidence. No information concerning the study, or the data will be released to any unauthorized third party.

The study monitor, representatives of the IRB, regulatory agencies or company supplying study product may inspect all documents and records required to be maintained by the investigator, including but not limited to, medical records (office, clinic, or hospital) and pharmacy records for the participants in this study. The clinical study site will permit access to such records.

The study participant's contact information will be securely stored at each clinical site for internal use during the study. At the end of the study, all records will continue to be kept in a secure location for as long a period as dictated by the reviewing IRB, institutional policies, or sponsor requirements.

Study participant research data, which is for purposes of statistical analysis and scientific reporting, will be transmitted to and stored at the Coordinating Center. This will not include the participant's contact or identifying information, unless otherwise specified in the informed consent form. Rather, individual participants and their research data will be identified by a unique study identification number. The study data entry and study management systems will be secured, and password protected. At the end of the study, study databases will be de-identified and may be made public.

7 References

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